

Lab Sheet 3

- Write a program to print the following using 'while loop'.
 - 'MATHEMATICS' 7 times.
 - first 20 even numbers
 - natural numbers from 15 to 25
 - cube of 'n' positive numbers from 5 onwards.
 - Sequence
 - 105, 98, 91, 7.
 - 2, 22, 222, 2222,n terms
 - sum of the digits of a number accepted from the user.
- Use the 'for loop' to print the following:
 - First 10 odd numbers
 - 3 to 16 and their cubes.
 - All prime numbers up to 'n'.
 - Full Multiplication Table of 'n'.
- Write a program to find the largest and smallest digit in a number.
- Write a program to count the total number of digits in a number and write it in reverse order.
- Write a program to print the sum of $1+2+6+24+120+\dots n$ terms.
- Write a program to print an approximate value of e^x , where x is an user input. You should not use Math modules. *Hint: Use Taylor series expansion* $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$
- Write a program to check whether a number is Armstrong or not.
- Armstrong number of the first kind is a number that is the sum of its own digits each raised to the power of the number of digits. For example, $153 = 1^3 + 5^3 + 3^3$.
- Print the following:

a.

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      *
     **
    ***
   ****
  *****
```

b.

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      *
     * *
    * * *
   * * * *
  * * * * *
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